Certificate complexity of uniform descent for the $3x + 1$ problem

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Abstract

We formalize the most natural proof strategy for the $3x+1$ (Collatz) problem — partition the integers into congruence classes and certify uniform descent on each class by an affine identity — as a proof system of dyadic affine descent certificates, and we prove quantitative lower bounds on its complexity. In every correct system the certificates deeper than k must occupy all $c(k)$ climbing classes mod 2^k, and truncation at depth K leaves uncovered density $c(K)/2^K$; $c(k)$ grows exactly like the binary entropy $H(\log_3 2) = 0.9500$ per digit, with sharp order $c(k) \asymp 2^{H(\log_3 2)k} k^{-3/2}$. The frontier is unchanged under disjunctions of congruence classes of arbitrary moduli and adaptive stopping times (Terras's coefficient stopping time), and an elementary count bounds the potential counterexamples up to N by $O(N^{H(\log_3 2)})$. The frontier is driven by the Sturmian word of slope $\log_3 2$ and is not regular. The counting theorem, the sandwich, and the core lemmas are machine-verified in Lean 4.

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1 Introduction

The $3x + 1$ problem (Collatz conjecture) asserts that iterating $C(n) = n/2$ for even n and $C(n) = 3n+1$ for odd n eventually reaches 1 from every positive integer. Despite its elementary statement, the problem has resisted every known technique; see the surveys of Lagarias [3, 4] for its history. Verified convergence now extends to all $n < 2^{68}$ [6]; almost every integer (in natural density) descends below itself, by Terras [1] and Everett [2]; and Tao [5] proved that almost all orbits (in logarithmic density) attain almost bounded values. Concurrently with and independently of the present work, Sharpe [21] has machine-verified in Lean 4 a no-go theorem for *bounded-window* descent certificates together with Terras's theorem in finite form; the relation to our results is discussed in Remarks 5.1, 6.6 and 8.2.

Faced with such a problem, the strategy that nearly every newcomer — and many published attempts — adopt is the same folk recipe: split the integers into congruence classes mod 2^k, prove on each class an affine identity forcing descent below the starting point, and conclude by strong

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induction. The recipe succeeds on a set of classes of density approaching 1, and this is precisely the content of the classical density theorems. However, the recipe never seems to terminate: no matter how deep the case analysis is pushed, a residue of unresolved classes survives, and it has remained unquantified *whether this is an artifact of particular implementations or a provable property of the strategy itself.*

Here we prove that it is a provable property of the strategy, and we determine its exact cost. We formalize the recipe as a proof system of *dyadic affine descent certificates* (Definition 3.1), flexible enough to cover finite exceptional sets, refinements by odd moduli, and arbitrary mixtures of depths, and we establish the following.

(i) *A frontier-width lower bound against all systems* (Theorem A): in every correct certificate system, the certificates of dyadic depth exceeding k must occupy *every* one of the $c(k)$ *climbing classes* mod 2^k, and any family of certificates of depth $\le K$ leaves uncovered a set of density $c(K)/2^K$. In particular no finite certificate system exists, and the cost of covering all but density ε is depth $\log_2(1/\varepsilon)/(1-H(\theta))$ up to lower order. The bound is robust (Theorem A′): allowing each certificate to certify descent at a *variable* stopping time, over a *disjunction* of congruence classes of arbitrary moduli, leaves the frontier unchanged — adaptive stopping collapses to the canonical first-descent time (the horizon-bounded form of Terras's coefficient stopping time), and each constituent class can serve at most one climbing class.

(ii) *The exact growth rate* (Theorem B): the non-asymptotic sandwich $\binom{k}{\lceil\theta k\rceil}/k \le c(k) \le (k+1)\binom{k}{\lceil\theta k\rceil}$, $\theta = \log_3 2$, whence $\gamma = \lim_k \frac{1}{k}\log_2 c(k) = H(\theta) = 0.9499555272\ldots$ The lower bound rests on an elementary rotation lemma of independent interest, and a fluctuation-theory argument shows it is sharp up to constants: $c(k) \asymp \binom{k}{\lceil\theta k\rceil}/k$, i.e. $\log_2 c(k) = H(\theta)k - \frac{3}{2}\log_2 k + O(1)$ (Theorem 7.4).

(iii) *Fine structure* (Theorems C–E): the climbing condition factorizes over the Sturmian word of slope θ; block transfer matrices obey the standard-word recursion governed by the continued fraction of $\log_2 3$; and the language of climbing words is not regular, so the certificate frontier admits no finite-automaton compression.

(iv) *Machine verification*: Theorems A and B are verified in Lean 4 — A in an abstract certificate-system formulation (as an injection from climbing classes into deep certificates), the two-sided binomial sandwich of B against Mathlib — and so are the core arithmetic lemmas behind Theorem A′ (the climbing towers $-1 \bmod 2^m$, the affine identity with its remainder bound, both halves of the adaptive collapse lemma, and the persistence step), all from the standard axioms alone (Section 10).

(v) *A verified computational artifact*: a depth-20 certificate system (4404 certificates, exact cover of $\mathbb{N}\setminus\{1\}$ up to the climbing classes, density 97.39%) checked by an independent symbolic verifier.

(vi) *Local obstructions* (Theorems F–G): no bounded syndrome correction $\psi(n \bmod 2^d)$ makes $\log_2 n + \psi$ a one-step Lyapunov function, and finite-window constraint hierarchies cannot exclude climbing statistics, because the all-ones statistics are realized by genuine integer orbit segments at every window length.

(vii) *A power-saving counterexample count* (Corollary 6.5): unconditionally, the number of $n \le N$ whose orbit never descends below n is $O(N^{H(\theta)}/\sqrt{\log N})$, by an elementary argument: the counting exponent is the certificate growth rate $H(\theta) = 0.9500$ itself.

We stress an explicit non-claim. The density exponent $1 - H(\theta)$ of the unresolved classes is classical: it is implicit in Terras's stopping time distribution [1] and appears in the survey literature [3, 16, 11]. What we claim as new is the *proof-complexity* statement: the transfer of the count from one greedy tree to *all* correct certificate systems (Theorem A), the two-sided non-asymptotic constants and the elementary proof of the exact rate (Theorem B), the Sturmian and automata-theoretic analysis of the frontier (Theorems C–E), the machine-verified core, and the verified depth-20 artifact. In the language of proof complexity, Theorems A–B are a lower bound for a natural, restricted proof system for the $3x+1$ problem, in the spirit of lower bounds for resolution or bounded-depth systems, and they rigorously prune a region of the strategy space in which a large fraction of amateur and computer-assisted attempts operate.

Related recent work. The parity-vector viewpoint has seen recent activity close to our tools: Rozier and Terracol [17] study the *paradoxical* orbit segments (affine coefficient below one, yet no descent) which are exactly the exceptional integers of our collapse lemma; Niu [18] enumerates paradoxical windows in closed form and observes their continued-fraction structure; and Fernández and Ibáñez [19] prove that Christoffel words are extremal for a cycle functional. We indicate the points of contact as they arise (Remarks 6.6 and 8.2); the proof-system lower bounds and the counting bound proved here do not appear in these works.

The paper is organized as follows. Section 2 fixes notation and proves the Terras bijection in the form we need. Section 3 defines certificates and systems. Section 4 proves the core lemmas, Section 5 the counting theorem, Section 6 its extension to disjunctive and adaptive systems, and Section 7 the exact growth rate. Section 8 develops the Sturmian structure, Section 9 the local obstructions, Section 10 describes the verified artifacts, and Section 11 delimits what is *not* blocked by these results. An appendix reproduces, for completeness, the standard cycle bound in the form used here.

2 Preliminaries: the compressed map and parity vectors

Throughout, T denotes the *compressed* Collatz map on the positive integers,

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+1)/2, & n \text{ odd}, \end{cases}$$

introduced by Terras [1]. For odd n one has $C^2(n) = T(n)$, and since $3n + 1 > n$, the orbit of n under C descends below n if and only if the orbit under T does; all our statements therefore transfer to the standard map by this time change. We write $\theta = \log_3 2 = 0.6309297536\ldots$ and $H(x) = -x \log_2 x - (1-x)\log_2(1-x)$ for the binary entropy.

For $n \ge 1$ and $j \ge 0$ let $b_j(n) = T^j(n) \bmod 2$ be the *parity vector* of the orbit, and $e_j(n) = b_0(n) + \cdots + b_{j-1}(n)$ the number of odd steps among the first j.

Lemma 2.1 (affine form and Terras bijection)**.** *For every $k \ge 0$ and every residue r mod 2^k:*

(a) the parities $b_0, \ldots, b_{k-1}$ are constant on the class $\{n \equiv r \pmod{2^k}\}$; write $e_j = e_j(r)$ for the resulting prefix counts;

(b) there are integers $\beta_j \geq 0$ $(0 \leq j \leq k)$, depending only on r, with

$$T^j(n) = \frac{3^{e_j} n + \beta_j}{2^j} \qquad (n \equiv r \bmod 2^k,\ 0 \leq j \leq k),$$

where $\beta_0 = 0$ and $\beta_{j+1} = 3\beta_j + 2^j$ if $b_j = 1$, $\beta_{j+1} = \beta_j$ if $b_j = 0$; moreover $\beta_j \leq 3^j$;

(c) the map $r \mapsto (b_0, \ldots, b_{k-1})$ is a bijection from $\mathbb{Z}/2^k\mathbb{Z}$ onto $\{0,1\}^k$.

Proof. Induction on k, all three claims jointly. For $k = 0$ there is nothing to prove. Assume the claims for k and fix a class r mod 2^k, with affine form $T^k(n) = (3^{e_k} n + \beta_k)/2^k$. If n and $n' = n + 2^k$ are the two lifts of r mod 2^{k+1}, then $T^k(n') - T^k(n) = 3^{e_k} \cdot 2^k / 2^k = 3^{e_k}$, which is odd. Hence b_k is constant on each class mod 2^{k+1} and takes different values on the two lifts of each class mod 2^k; this proves (a) and (c) at level $k+1$. For (b), if $b_k = 0$ then $T^{k+1}(n) = T^k(n)/2 = (3^{e_k} n + \beta_k)/2^{k+1}$; if $b_k = 1$ then $T^{k+1}(n) = (3\,T^k(n) + 1)/2 = (3^{e_k+1} n + 3\beta_k + 2^k)/2^{k+1}$. The update rules and $\beta_{j+1} \leq 3\beta_j + 2^j \leq 3 \cdot 3^j$ give $\beta_j \leq 3^j$ by induction. □

3 Certificates and systems

Definition 3.1 (dyadic affine descent certificate)**.** A *certificate* is a tuple (r, k, e, β, E) where $k \geq 1$, r is a residue mod 2^k whose parity vector has $e = e_k(r)$ odd steps and affine constant $\beta = \beta_k(r)$ as in Lemma 2.1, E is a finite set, and the *descent condition*

$$T^k(n) < n \qquad \text{for all } n \in \mathcal{C}, \qquad \mathcal{C} = \{n \geq 1 : n \equiv r \bmod 2^k\} \setminus E,$$

holds. More generally we allow the certificate to be restricted to a class $\mathcal{C}' \subseteq \mathcal{C}$ defined by an additional congruence to an odd modulus. In all cases the *dyadic atom* of the certificate is the class r mod 2^k in which it lives, and k is its *dyadic depth*.

A *certificate system* S is a finite or infinite family of certificates, together with a finite set D of directly verified integers, such that the union of the certificate classes and D covers $\mathbb{N} \setminus \{1, 2\}$.

A certificate system is exactly what is needed to run the folk proof: if every $n \geq 3$ lies in some certified class (or in D, whose orbits are checked to reach 1 directly), strong induction on n proves the conjecture, provided the exceptional sets are also checked. Whether a correct system exists at all is therefore equivalent to the conjecture itself (if the conjecture holds, the canonical first-descent tree is one); the point of this paper is that *any* correct system must be infinite, with a quantified, exponentially growing frontier at every depth. The verified artifact of Section 10 is a finite *partial* system: it certifies all classes mod 2^{20} except the $c(20)$ climbing ones.

4 Climbing classes and the core lemmas

Definition 4.1. A residue r mod 2^k (equivalently, by Lemma 2.1(c), its parity vector) is *climbing* if $3^{e_j} > 2^j$ for every $1 \leq j \leq k$; equivalently $e_j > \theta j$ for all $j \leq k$. Let $c(k)$ denote the number of climbing residues mod 2^k, and let $L \subseteq \{0,1\}^*$ be the language of climbing words.

The first lemma shows that climbing classes exist in abundance — indeed the class -1 mod 2^m climbs for $m - 1$ full steps, realizing the extremal growth rate $(3/2)^{m-1}$.

Lemma 4.2 (climb lemma; machine-verified)**.** *Let $m \geq 1$, $t \geq 1$ and $n = 2^m t - 1$. Define $n_j = 3^j 2^{m-j} t - 1$ for $0 \leq j \leq m-1$, so $n_0 = n$. Then for $0 \leq j \leq m-2$: n_j is odd, $T(n_j) = n_{j+1}$, and $n_{j+1} > n_j$. In particular the first $m-1$ steps of the orbit of n are all odd steps, the orbit is strictly increasing along them, and $n_j \equiv -1 \pmod{2^{m-j}}$ for all $j \leq m-1$.*

Proof. For $j \leq m-1$ the number $3^j 2^{m-j} t$ is even (as $m - j \geq 1$), so n_j is odd. For $j \leq m-2$,

$$T(n_j) = \frac{3n_j + 1}{2} = \frac{3^{j+1} 2^{m-j} t - 2}{2} = 3^{j+1} 2^{m-j-1} t - 1 = n_{j+1},$$

and $T(n) = (3n+1)/2 > n$ holds for every positive n. The congruence is read off the formula. This lemma is machine-verified in Lean 4 (Section 10). □

Corollary 4.3. *For every $k \geq 1$ the class -1 mod 2^k is climbing; hence $c(k) \geq 1$.*

Proof. By Lemma 4.2 (with $m = k$), the values $n_0, \ldots, n_{k-1}$ are all odd, so the parity vector of the class is all-ones: $e_j = j$ and $3^j > 2^j$ for every $j \leq k$. □

Lemma 4.4 (climbing classes admit no certificate)**.** *If $3^e \geq 2^k$ and $\beta \geq 0$, then $(3^e n + \beta)/2^k < n$ holds for no $n \geq 1$. Consequently a certificate of depth k (which certifies descent for all but finitely many n in an infinite class) must have $3^{e_k} < 2^k$, i.e. its dyadic atom is non-climbing at its own depth.*

Proof. $(3^e n + \beta)/2^k \geq n \cdot 3^e / 2^k \geq n$. □

Lemma 4.5 (prefix closedness and locality)**.** *(a) Climbing is prefix-closed: if r mod 2^k is climbing and $k' \leq k$, then the reduction r mod $2^{k'}$ is climbing.*

(b) A certificate of dyadic depth $k' \leq k$ covers no element of any climbing class mod 2^k.

(c) A certificate of dyadic depth $k'' > k$ has its atom contained in exactly one class mod 2^k.

Proof. (a) is immediate from Definition 4.1, since the conditions for $j \leq k'$ are a subset of those for $j \leq k$ and, by Lemma 2.1, the parity prefix of the reduced class is the prefix of the parity vector. For (b): by Lemma 4.4 the certificate's atom r' mod $2^{k'}$ satisfies $3^{e_{k'}} < 2^{k'}$, so it is not climbing at depth k'; but by (a) the reduction mod $2^{k'}$ of any climbing class mod 2^k *is* climbing at depth k'; hence the two classes mod $2^{k'}$ are distinct and therefore disjoint. (c) is trivial. □

5 The counting theorem

Theorem A (frontier-width lower bound). Let S be any correct certificate system. Then for every $k \geq 1$ the certificates of S of dyadic depth exceeding k have atoms lying in every climbing class mod 2^k — hence in at least $c(k)$ distinct classes mod 2^k. Consequently:

(a) any family of certificates all of dyadic depth $\leq K$ leaves every climbing class mod 2^K uncovered, a set of natural density $c(K)/2^K$; in particular no finite certificate system exists;

(b) with Theorem B below, the frontier of every correct system at depth k has width $c(k) = 2^{H(\theta)k + o(k)}$, and covering all but density ε requires depth $K \geq \log_2(1/\varepsilon)/(1 - H(\theta) + o(1))$.

We stress that the count is a statement about *distinct classes*: in a correct system the number of certificates deeper than k is infinite for every k (a finite number would bound all depths, which (a) forbids), so the content of the theorem is the width of the frontier, not a cardinality.

Proof. Fix k and a climbing class $\mathcal{K}$ mod 2^k. The class contains infinitely many integers and only finitely many lie in the directly verified set D, so some $n \in \mathcal{K}$ is covered by a certificate of S. By Lemma 4.5(b), no certificate of depth $\le k$ covers any element of $\mathcal{K}$; hence that certificate has depth $> k$. By Lemma 4.5(c), a certificate of depth $> k$ meets exactly one class mod 2^k, so certificates assigned in this way to distinct climbing classes are distinct. There are $c(k)$ climbing classes, each met by its own certificate of depth $> k$. Claim (a): by Lemma 4.5(b) no certificate of depth $\le K$ covers any element of a climbing class mod 2^K, and the climbing classes have density $c(K)/2^K \ge 2^{-K}$ (Corollary 4.3); a finite system has a maximal depth. Claim (b) follows from Theorem B and Corollary 7.6. □

Remark 5.1. Theorem A is machine-verified in an abstract certificate-system formulation, with the count rendered as an injection from climbing classes into deep certificates (Section 10). Sharpe's no-go theorem [21] excludes, by the same all-ones mechanism (the integers $2^L - 1$ shadow the 2-adic fixed point -1), every scheme whose potential is $n\,W(n)$ with W bounded above and below and whose descent window is uniformly bounded — the regime of a *finite* computation. Theorem A addresses the complementary regime of unbounded depth and quantifies it: the frontier width $c(k) = 2^{H(\theta)k+o(k)}$, its robustness under disjunction and adaptive stopping (Theorem A′), and its Sturmian and non-regular structure (Theorems C–E) have no counterpart in the bounded-window statement. The proof uses nothing about the internal structure of the certificates beyond Lemmas 4.4 and 4.5; in particular it applies verbatim to systems mixing arbitrary depths, refining by odd moduli, and discarding arbitrary finite exceptional sets. Systems whose members certify descent for a *disjunction* of congruence classes under an *adaptive* stopping rule are not covered by the statement as given; Section 6 extends the bound to exactly that setting.

6 Robustness: disjunctions and adaptive stopping

The folk strategy admits two natural liberalizations that Definition 3.1 does not cover: a single certificate might claim descent for a *union* of congruence classes, and it might allow the descent time to *vary with* n instead of being the fixed depth of an affine identity. This section shows that neither liberalization, nor both together, weakens the counting bound: adaptive stopping collapses to a canonical stopping time on every congruence class, and each constituent class of a valid disjunction can serve at most one climbing class.

Definition 6.1 (adaptive disjunctive certificate)**.** An *adaptive disjunctive certificate* is a tuple (U, K, E) where $U = C_1 \cup \cdots \cup C_w$ is a finite union of congruence classes $C_t = \{n \ge 1 : n \equiv r_t \bmod 2^{d_t} q_t\}$ with q_t odd (we call d_t the *dyadic depth* of the constituent C_t, and w the *width*), $K \ge 1$ is the *horizon*, E is finite, and the claim

$$\forall\, n \in U \setminus E \;\; \exists\, j \le K : \; T^j(n) < n$$

is true. An *adaptive system* is a countable family of such certificates, plus a finite directly verified set D, with $\bigcup_i (U_i \setminus E_i) \cup D \supseteq \mathbb{N} \setminus \{1, 2\}$. Since every eventually periodic subset of $\mathbb{N}$ is, up to a finite set, a finite union of congruence classes, this notion covers every certificate whose scope

is defined by congruence information of any modulus — including odd moduli such as powers of 3.

Lemma 6.2 (adaptive collapse)**.** *Fix a class $r \bmod 2^K$ with prefix counts e_j (Lemma 2.1) and let $j_1 = \min\{j \le K : 3^{e_j} < 2^j\}$ (with $j_1 = \infty$ if the class is climbing at depth K). Then for every $n \equiv r \pmod{2^K}$:*

(a) $T^j(n) > n$ for every $j \le K$ with $3^{e_j} > 2^j$ — with no condition on n; this holds in particular for every $j < j_1$, and if $j_1 = \infty$ then no $n \ge 1$ in the class descends below itself within K steps;

(b) if $j_1 \le K$ and $n > 3^{j_1}$ then $T^{j_1}(n) < n$.

Consequently, for $n > 3^K$ the events "$T^j(n) < n$ for some $j \le K$" and "$j_1 \le K$" coincide, and when $j_1 \le K$ the first descent time equals j_1: on every congruence class, an adaptive stopping time collapses to the canonical first-descent time. (This is the elementary, horizon-bounded direction of Terras's coefficient stopping time analysis [1].)

Proof. (a) $T^j(n) = (3^{e_j} n + \beta_j)/2^j \ge 3^{e_j} n/2^j > n$ whenever $3^{e_j} > 2^j$, with no condition on n; for $j < j_1$ this inequality holds by minimality of j_1. (b) Descent at j_1 is $n(2^{j_1} - 3^{e_{j_1}}) > \beta_{j_1}$; the left side is $\ge n$ (the bracket is a positive integer) and $\beta_{j_1} \le 3^{j_1}$ by Lemma 2.1(b). The consequence follows since steps $j < j_1$ never descend by (a) and step j_1 descends by (b). Both halves, together with the affine identity and the remainder bound $\beta_j + 2^j \le 3^j$, are machine-verified in Lean 4 (Section 10). □

Lemma 6.3 (persistence of climbing)**.** *If $r \bmod 2^k$ is climbing, its extension appending the parity 1 is a climbing class mod 2^{k+1}. Hence $c(k+1) \ge c(k)$; every climbing class mod 2^k contains a climbing class mod 2^K for every $K \ge k$; and the intersection of a climbing class with any congruence class to an odd modulus is infinite.*

Proof. Prefix conditions up to k are unchanged, and at $k+1$: $3^{e_k+1} = 3 \cdot 3^{e_k} > 3 \cdot 2^k > 2^{k+1}$. Iterating gives the second claim; the third is the Chinese remainder theorem. The arithmetic step is machine-verified (Section 10). □

Theorem A′ (robustness under disjunction and adaptive stopping). Let S be any correct adaptive system. Then:

(a) every constituent class of every certificate of S is non-climbing at its own dyadic depth;

(b) for every $k \ge 1$, the constituent classes of dyadic depth exceeding k occupy every climbing class mod 2^k (hence at least $c(k)$ distinct classes), and every certificate covering an element of a climbing class mod 2^k has horizon exceeding k;

(c) (packaging invariance) for every valid (U, K, E),

$$U \setminus \big(E \cup [1, 3^K]\big) \;\subseteq\; D(K) := \bigcup\{\,\text{classes mod } 2^K \text{ with } j_1 \le K\,\},$$

and conversely $D(K)$ is covered by the canonical uniform tree of first-descent certificates of depth $\le K$ with exceptions in $[1, 3^K]$. Every adaptive disjunctive system is therefore, up to finite sets, a repackaging of the canonical first-descent tree.

Proof. (a) Suppose the constituent C_t (dyadic depth d, odd modulus q_t, inside a certificate of horizon K) is climbing at depth d. By Lemma 6.3 there is a climbing class mod $2^{\max(d,K)}$ inside its dyadic class, and its intersection with the congruence condition mod q_t is infinite; every n in that intersection satisfies $T^j(n) > n$ for all $j \le K$ by Lemma 6.2(a). All but finitely many of these n avoid E, so the certificate's claim is false.

(b) Fix a climbing class $\mathcal{K}$ mod 2^k and pick $n \in \mathcal{K} \setminus D$ covered by some certificate, say $n \in C_t \setminus E$. By (a) the class C_t has a first non-climbing prefix $j_1 \le d_t$. Since $n \in \mathcal{K}$, the parity prefix of n is climbing at every depth $\le k$; since $n \in C_t$, its prefix at depth j_1 is that of C_t, which is non-climbing. Hence $j_1 > k$, so $d_t \ge j_1 > k$, and the dyadic class of C_t reduces mod 2^k to the single class $\mathcal{K}$. Distinct climbing classes therefore meet distinct constituent classes, so the constituents of dyadic depth $> k$ occupy every climbing class mod 2^k. For the horizon: the prefixes of n at depths $\le k$ are climbing, so $T^j(n) > n$ for all $j \le k$ (Lemma 6.2(a)), and the certified descent time of n exceeds k; hence $K > k$.

(c) For $n \in U \setminus E$ with $n > 3^K$, the claim provides a descent within K steps, and Lemma 6.2 shows this forces $j_1(n \bmod 2^K) \le K$, i.e. $n \in D(K)$. Conversely each class mod 2^K with $j_1 \le K$ is certified by the affine certificate of depth j_1 at its first-descent prefix, whose exceptional set is contained in $[1, 3^{j_1}]$ by Lemma 6.2(b). □

Remark 6.4 (numerical verification)**.** The collapse of Lemma 6.2 was checked exhaustively for all $2 \le n \le 2^{20}$ at horizon $K = 12$ (no violations; in this range no exceptional stragglers below the 3^{j_1} bound occurred at all), the monotonicity $c(k+1) \ge c(k)$ was confirmed to $k = 20$ with the anchor $c(20) = 27328$ of Section 10, and invalidity witnesses for climbing constituents with odd moduli $q \in \{5, 7, 11\}$ were confirmed at horizons up to 80.

The collapse lemma has a further unconditional consequence: a power-saving upper bound on the number of potential counterexamples to the conjecture itself.

Corollary 6.5 (counting potential counterexamples)**.** *Let $\mathcal{E} = \{n > 1 : T^j(n) \ge n$ for all $j \ge 1\}$, the set of integers whose orbit never descends below its starting point. (The least element of any divergent orbit and of any nontrivial cycle lies in $\mathcal{E}$, and the conjecture is equivalent to $\mathcal{E} = \emptyset$.) Then, unconditionally and with an absolute constant C, for all $N \ge 2$,*

$$|\mathcal{E} \cap [1, N]| \;\le\; C\,\frac{N^{\,H(\theta)}}{\sqrt{\log N}}, \qquad H(\theta) = 0.9499555\ldots :$$

the counting exponent equals the certificate growth rate γ.

Proof. Let $n \in \mathcal{E} \cap (M, 2M]$ with $M = 2^m$, and let e be the number of odd steps among the first m steps of the orbit of n. Every iterate n' is $\ge n > M$, so an odd step multiplies the current value by $(3n'+1)/(2n') \le (3 + 1/M)/2$ and an even step by $1/2$; hence

$$1 \;\le\; \frac{T^m(n)}{n} \;\le\; 2^{-m}\Big(3 + \frac{1}{M}\Big)^e, \qquad \text{i.e.} \qquad e \;\ge\; m\,\theta_M, \quad \theta_M := \frac{\log 2}{\log(3 + 1/M)} = \theta - O(1/M).$$

By Lemma 2.1(c), each parity vector of length m occurs for exactly one integer of $(M, 2M]$, so, with $e_1 = \lceil m\theta_M \rceil$,

$$|\mathcal{E} \cap (M, 2M]| \le \sum_{e \ge e_1} \binom{m}{e} \le \frac{1}{1-\rho}\binom{m}{e_1} \le C_0\,\frac{2^{mH(\theta_M)}}{\sqrt{m}},$$

where $\rho = (m-e_1)/(e_1+1) \le (1-\theta_M)/\theta_M < 0.6$ bounds the ratio of consecutive binomial coefficients beyond $e_1 > m/2$, and the last step is Stirling's bound $\binom{m}{e} \le C\,2^{mH(e/m)}m^{-1/2}$ for e/m in a compact subinterval of $(0,1)$. As $m\,(H(\theta_M)-H(\theta)) = O(m/M) \to 0$, this is at most $C_0'\,M^{H(\theta)}(\log_2 M)^{-1/2}$ with an absolute C_0', and summing over the dyadic blocks $M = 2^m \le N$ (a geometric sum dominated by its largest term) gives the claim. □

Remark 6.6. (i) The congruence-shadow route gives the same exponent with more effort: Lemma 6.2 forces every $n \in \mathcal{E}$ to be climbing at all depths $j \le \log_3 n$, which yields the weaker exponent $1-\theta(1-H(\theta)) = 0.9684254\ldots$; forcing climbing up to depth $\log_2 n$ (and hence $H(\theta)$ again) requires bounding the *margin* $B(j) := \max\lfloor \beta_j/(2^j - 3^{e_j})\rfloor$ over classes whose first non-climbing depth is j, e.g. through the effective irrationality measure $|j\log 2 - e\log 3| > c\,j^{-13.3}$ of Rhin [9]. The integers that dodge descent at their first non-climbing depth are initial terms of *paradoxical sequences* in the sense of Rozier–Terracol [17] — a proper subclass, since a paradoxical window may also end after a descent (for $n=7$ the coefficient first drops below one at $j=7$, where $T^7(7)=5<7$, yet the length-8 window with $T^8(7)=8$ is paradoxical). Their remainder bound [17, Thm. 2.4], $\beta_j/2^j \le (3^q-2^q)/2^q$ with $q=e_j$, bounds the margin by $(3/2)^{\theta j}j^{13.3}$ up to constants, and their Theorem 1.3 shows no paradoxical sequence starts in $(4614,\ 2.8\times10^{19})$. Numerically $B(j) \le 24$ for all $j \le 24$ (the real-valued maximum of $\beta_j/(2^j-3^{e_j})$ is 24.54 at $j=8$), with spikes at the depths where 2^j is unusually close to a power of 3 from above (e.g. $j=8$: $2^8/3^5 = 256/243$, the Pythagorean limma) — the same continued-fraction structure observed for paradoxical ratios in [18]. (ii) The argument of the corollary is Terras's [1]: the parity vector of length m is a bijection on $(M, 2M]$, and non-descent forces a supercritical density of odd steps. Terras's theorem in this finite form was recently machine-verified by Sharpe [21], with the explicit bound $\#\{n \in [2^k, 2^{k+1}) : \text{no descent within } k \text{ steps}\} \le 3^k/2^{\lfloor 17(k-1)/27\rfloor} \approx 2^{0.955k}$, where the rational approximation $3^{17} \le 2^{27}$ replaces the entropy function; the corollary sharpens the exponent to the exact entropy value $H(\theta) = 0.94996\ldots$ with an absolute constant (and a $(\log N)^{-1/2}$ saving from the binomial ratio bound) and identifies it with the certificate growth rate γ. The recent parity-vector literature enumerates paradoxical *windows* [18] and shows that finiteness of the set of paradoxical sequences implies the conjecture [17, Cor. 3.3]. We are not aware of a stated count of $\mathcal{E}$ with the exponent $H(\theta)$, but the bound is a routine consequence of Terras's method, and we state it with that caveat; its combinatorial core is machine-verified (Section 10). (iii) The shadow method is exhausted at the exponent $H(\theta)$: the set of integers in $(M, 2M]$ satisfying all depth-m congruence necessary conditions has exactly $c(m)$ elements, so no argument using congruence shadows alone can lower the exponent further. Pushing the forced-climbing depth to its true limit $\log_{3/2} n$ would require counting climbing residues that are *small as integers* — information not contained in shadows (heuristically that limit exponent is $1-(1-H(\theta))/\log_2(3/2) = 0.91445\ldots$). (iv) Thematically, the corollary ties the two halves of the problem together: the count of potential counterexamples is governed by the same entropy that prices the certificate frontier, and the finer margin analysis of (i) runs through the same linear-forms data that governs the cycle bound of Appendix A.

Remark 6.7 (what robustness does and does not say)**.** Theorem A′ closes every liberalization of the folk strategy whose certificate scopes are eventually periodic sets — in particular, partitioning by residues modulo powers of 3, or any mixed modulus, buys nothing, and neither does any orbit-adaptive stopping rule. What remains outside its quantification are certificates whose scopes are *not* unions of congruence classes: sets defined by inequalities on orbit values, by unbounded-window parity information, or by genuinely analytic (ensemble) data. Section 11 discusses these exits.

7 The exact growth rate

Lemma 7.1 (rotation lemma)**.** *Let $b = (b_1, \dots, b_k) \in \{0,1\}^k$ with $e = \sum_i b_i$ and $e - \theta k > 0$. Then some cyclic rotation of b is climbing, i.e. has all prefix sums $e_j - \theta j$ positive.*

Proof. Set $w_i = b_i - \theta$ and $S_j = w_1 + \cdots + w_j$, $S_0 = 0$, so $S_k = e - \theta k > 0$. Since θ is irrational, the values $S_0, \dots, S_{k-1}$ are pairwise distinct (an equality $S_a = S_b$ with $a < b$ would make $\theta = (e_b - e_a)/(b-a)$ rational). Let j^* be the unique index in $\{0, \dots, k-1\}$ at which $\min_{0 \le j \le k-1} S_j$ is attained, and consider the rotation $b' = (b_{j^*+1}, \dots, b_k, b_1, \dots, b_{j^*})$. Its prefix sums are

$$S'_i = S_{j^*+i} - S_{j^*} \quad (1 \le i \le k - j^*), \qquad S'_{k-j^*+i} = S_k - S_{j^*} + S_i \quad (1 \le i \le j^*).$$

The first group is positive because S_{j^*} is the strict minimum among $S_0, \dots, S_{k-1}$ and $S_k > S_0 \ge S_{j^*}$; the second group is positive because $S_i \ge S_{j^*}$ and $S_k > 0$. Hence all prefix sums of b' are positive, which is the climbing condition. □

Remark 7.2 (machine verification; irrationality is removable)**.** The lemma is machine-verified (Section 10) in a purely integer form: comparisons $S_a \le S_b$ are encoded as the cross-multiplied inequalities $3^{e_a} 2^b \le 3^{e_b} 2^a$, and the rotated word's prefix conditions become the two families of integer inequalities produced by the proof. The formalization selects the *last* index attaining the minimum, which yields the required strictness on the first group directly — so the appeal to the irrationality of θ in the proof above is a convenience of the real-variable formulation, not a genuine hypothesis.

Theorem B (exact growth rate). For every $k \ge 1$, with $\theta = \log_3 2$,

$$\frac{1}{k}\binom{k}{\lceil \theta k \rceil} \;\le\; c(k) \;\le\; (k+1)\binom{k}{\lceil \theta k \rceil}.$$

Consequently $\gamma := \lim_{k\to\infty} \frac{1}{k} \log_2 c(k)$ exists and

$$\gamma = H(\theta) = H(\log_3 2) = 0.9499555272\ldots$$

Moreover c is supermultiplicative, $c(k_1{+}k_2) \ge c(k_1)c(k_2)$, so the limit also equals $\sup_k \frac{1}{k} \log_2 c(k)$.

Proof. Lower bound. Let $e = \lceil \theta k \rceil$, so $e - \theta k > 0$ (θk is irrational). Partition the $\binom{k}{e}$ words with e ones into classes under cyclic rotation; each class has at most k elements, and by Lemma 7.1 each class contains at least one climbing word. By Lemma 2.1(c) every word is the parity vector of exactly one residue, so $c(k) \ge \binom{k}{e}/k$.

Upper bound. A climbing word satisfies $e_k > \theta k$, so $e_k \ge \lceil \theta k \rceil$. Since $\theta > 1/2$, the binomial coefficients $\binom{k}{e}$ are non-increasing for $e \ge \lceil \theta k \rceil \ge k/2$, whence

$$c(k) \le \sum_{e \ge \lceil \theta k \rceil} \binom{k}{e} \le (k + 1 - \lceil \theta k \rceil)\binom{k}{\lceil \theta k \rceil} \le (k+1)\binom{k}{\lceil \theta k \rceil}.$$

Supermultiplicativity. If u is climbing of length k_1 and v climbing of length k_2, then the concatenation uv is climbing: for $j \le k_1$ the prefix sums are those of u; for $j = k_1 + i$ they equal $S^u_{k_1} + S^v_i > 0$. Distinct pairs give distinct concatenations, so $c(k_1 + k_2) \ge c(k_1)c(k_2)$. Fekete's lemma applied to the superadditive sequence $\log_2 c(k)$ (finite since $c(k) \le 2^k$, and $c(k) \ge 1$ by Corollary 4.3) yields $\lim_k \frac{1}{k} \log_2 c(k) = \sup_k \frac{1}{k} \log_2 c(k)$.

The value. By Stirling, $\frac{1}{k} \log_2 \binom{k}{\lceil \theta k \rceil} \to H(\theta)$, and the polynomial factors $1/k$, $(k+1)$ do not affect the limit. □

Remark 7.3 (numerical verification)**.** The sandwich was verified computationally for all $k \le 1600$ against exact values of $c(k)$ computed by dynamic programming over the walk of Theorem C, and the climbing condition was checked against exhaustive enumeration of all 2^k words for $k \le 16$. The fluctuation $\tilde{b}(k) = \log_2 c(k) - \log_2 \binom{k}{\lceil \theta k \rceil}$ stays within the proven window $[-\log_2 k,\ \log_2(k+1)]$ throughout.

Theorem 7.4 (sharp growth)**.** *There are absolute constants $0 < c_1 \le c_2 < \infty$ such that for all $k \ge 1$*

$$c_1 \frac{1}{k}\binom{k}{\lceil \theta k \rceil} \ \le\ c(k)\ \le\ c_2 \frac{1}{k}\binom{k}{\lceil \theta k \rceil}, \qquad \textit{hence} \qquad \log_2 c(k) = H(\theta)\, k - \tfrac{3}{2}\log_2 k + O(1).$$

More precisely, writing $N_k(e)$ for the number of climbing words of length k with e ones and $D' = e - \theta k$, one has $N_k(e) \le c_2 \binom{k}{e}(1 + D')/k$ whenever $0 < D' \le \sqrt{k}$.

The lower bound is Theorem B (with $c_1 = 1$); the content is the upper bound, which says that the rotation lemma is sharp on average: a cyclic class of words with e ones contains $O(1 + D')$ climbing rotations, not $O(k)$. The proof is a ballot estimate for a Bernoulli bridge against a line of irrational slope, obtained from fluctuation theory; sharp asymptotics for such conditioned local probabilities are known in far greater generality [24, 25], but since only upper bounds are needed we give a short self-contained argument whose single imported fact is the harmonicity of the ladder-height renewal function [22, 23].

Proof. Let the letters of a word of length k be i.i.d. with $\mathbb{P}(1) = \theta$, let B_j be the number of ones among the first j letters, and set $S_j := B_j - \theta j$. Then (S_j) is a mean-zero random walk with steps $1 - \theta$ (probability θ) and $-\theta$ (probability $1 - \theta$), and the word is climbing iff $S_j > 0$ for $1 \le j \le k$, i.e. iff $\tau > k$ where $\tau := \min\{j \ge 1 : S_j \le 0\}$. Since θ is irrational, $S_j \ne 0$ for $j \ge 1$, and more generally $S_j \ne S_i$ for $i < j$. Conditionally on $B_k = e$ the word is uniform among words with e ones, so

$$N_k(e) = \binom{k}{e} \frac{\mathbb{P}(\tau > k,\ B_k = e)}{\mathbb{P}(B_k = e)}. \tag{1}$$

For $y \ge 0$ write $\mathbb{P}_y$ for the walk started at $S_0 = y$ (i.e. $S_j = y + B_j - \theta j$), with the same τ. We use three facts.

(F1) Local bounds for the binomial. There are constants $c_0, C_0 > 0$ with $\sup_z \mathbb{P}(B_m = z) \le C_0 m^{-1/2}$ for all $m \ge 1$, and $\mathbb{P}(B_k = e) \ge c_0 k^{-1/2}$ whenever $|e - \theta k| \le \sqrt{k}$ (Stirling). The same holds for the number of zeros, hence for any walk whose increments take two values with fixed probabilities.

(F2) Harmonic renewal function. Let $H_1, H_2, \ldots$ be the descending ladder heights of (S_j) under $\mathbb{P}_0$ (the successive amounts by which new minima undercut old ones; strict and weak ladder heights coincide here because the walk never revisits a level). Each $H_i \in (0, \theta]$, since the walk descends only by steps of size θ, and $\mu := \mathbb{E}H_1 > 0$. The renewal function $V(y) := \sum_{i \ge 0} \mathbb{P}(H_1 + \cdots + H_i \le y)$ satisfies the classical identity $V(y) = \mathbb{E}_y[V(S_1);\ S_1 > 0]$ for $y > 0$ [22, 23], hence by iteration $V(y) = \mathbb{E}_y[V(S_n);\ \tau > n]$ for all n. Moreover $y \le V(y) \le (y + 1)/\mu$: the lower bound because the ladder sums increase by at most 1 per step, the upper bound by Wald's identity applied to the first ladder sum exceeding y.

(F3) Uniform persistence bound. We claim $u_n(y) := \mathbb{P}_y(\tau > n) \le C_1(1 + y)n^{-1/2}$ for all $n \ge 1$, $y \ge 0$. Fix $\varepsilon \in (0, 1)$, $n \ge 2$, $y > 0$, and $m = \lfloor n/2 \rfloor$. By (F2) and Markov's inequality,

since V is non-decreasing and $V(x) \geq x$,

$$\mathbb{P}_y(\tau > n,\ S_n > \varepsilon\sqrt{n}) \leq \frac{\mathbb{E}_y[V(S_n); \tau > n]}{V(\varepsilon\sqrt{n})} \leq \frac{V(y)}{\varepsilon\sqrt{n}} \leq \frac{(1+y)}{\mu\,\varepsilon\sqrt{n}}.$$

On the complementary event, by the Markov property at time m and (F1), $\mathbb{P}_y(\tau > n,\ S_n \leq \varepsilon\sqrt{n}) \leq u_m(y) \cdot \sup_x \mathbb{P}\big(x + S_{n-m} \in (0, \varepsilon\sqrt{n}]\big) \leq u_m(y) \cdot C_0(\varepsilon\sqrt{n}+1)(n-m)^{-1/2}$, because S_{n-m} lives on a translate of $\mathbb{Z}$ and the interval contains at most $\varepsilon\sqrt{n}+1$ lattice points. Choosing $\varepsilon = 1/(8C_0)$ and $n \geq n_0 := (8C_0)^2$ gives $u_n(y) \leq \frac{1}{2}u_{\lfloor n/2 \rfloor}(y) + C(1+y)n^{-1/2}$, and iterating $j = \lfloor \log_2(n/n_0) \rfloor$ times yields $u_n(y) \leq 2^{-j} + C(1+y)n^{-1/2}\sum_{i \geq 0} 2^{-i/2} \leq C_1(1+y)n^{-1/2}$. The case $y = 0$ follows by conditioning on the first step ($S_1 > 0$ forces $S_1 = 1-\theta$), and $n < n_0$ is trivial. The same bound holds for the *reversed* walk $\hat{S}$ with steps θ (probability $1-\theta$) and $-(1-\theta)$ (probability θ), whose ladder heights lie in $(0, 1-\theta]$.

The bridge bound. Let $0 < D' = e - \theta k \leq \sqrt{k}$ and $m = \lfloor k/2 \rfloor$, $n = k - m$. By the Markov property at time m, $\mathbb{P}(\tau > k, B_k = e) = \sum_z \mathbb{P}(\tau > m, B_m = z)\,\mathbb{P}_{y_z}(\tau > n,\ S_n = D')$ with $y_z = z - \theta m > 0$. Reading the last n letters backwards turns a path from y_z to D' that stays positive into a path of the reversed walk from D' to y_z that stays positive; since the letters are i.i.d., $\mathbb{P}_{y_z}(\tau > n, S_n = D') = \hat{\mathbb{P}}_{D'}(\hat{\tau} > n, \hat{S}_n = y_z)$. Splitting once more at time $\lfloor n/2 \rfloor$ and using (F3) for $\hat{S}$ and then (F1),

$$\hat{\mathbb{P}}_{D'}(\hat{\tau} > n, \hat{S}_n = y_z) \leq \hat{\mathbb{P}}_{D'}(\hat{\tau} > \lfloor n/2 \rfloor) \cdot \sup_x \hat{\mathbb{P}}(\hat{S}_{n-\lfloor n/2 \rfloor} = x) \leq \frac{C_1(1+D')}{\sqrt{n/2}} \cdot \frac{C_0}{\sqrt{n/2}} \leq \frac{C(1+D')}{k}.$$

Summing over z and using (F3) once more, $\mathbb{P}(\tau > k, B_k = e) \leq C(1+D')k^{-1}\mathbb{P}(\tau > m) \leq C'(1+D')k^{-3/2}$. With (1) and the lower bound in (F1) this gives $N_k(e) \leq c_2\binom{k}{e}(1+D')/k$.

Summation. Let $e_0 = \lceil \theta k \rceil$ and $e = e_0 + j$. Since $e_0 \geq \theta k$, $\binom{k}{e+1}/\binom{k}{e} = (k-e)/(e+1) \leq (1-\theta)/\theta =: \rho < 0.6$, so $\binom{k}{e_0+j} \leq \rho^j\binom{k}{e_0}$, and $D'_e \leq j+1$. Hence

$$c(k) = \sum_{e \geq e_0} N_k(e) \leq \frac{c_2}{k}\binom{k}{e_0}\sum_{j \geq 0}(j+2)\rho^j + \binom{k}{e_0}\sum_{j > \sqrt{k}}\rho^j \leq C\,\frac{1}{k}\binom{k}{e_0},$$

the tail being $O(\rho^{\sqrt{k}}) = O(1/k)$. Finally $\binom{k}{\lceil \theta k \rceil} = 2^{H(\theta)k}k^{-1/2}\,2^{O(1)}$ by Stirling, since $k\,|H(\lceil \theta k \rceil/k) - H(\theta)| = O(1)$. □

Remark 7.5. Numerically $R(k) := k\,c(k)/\binom{k}{\lceil \theta k \rceil}$ lies in $[11.6,\ 20.7]$ for $200 \leq k \leq 1200$ (exact values), with no trend in k (the regression slope of $\log_2 R$ on $\log_2 k$ is 0.07) and a monotone dependence on the offset $\lceil \theta k \rceil \log_2 3 - k \in (0, \log_2 3)$, as the $(1+D')$ factor predicts; the finer ratios $kN_k(e)/\big(\binom{k}{e}(1+D')\big)$ stay in $[0.91, 2.12]$. The three auxiliary bounds (F1)–(F3) and the bridge bound were also checked by exact dynamic programming (supplementary scripts). A least-squares fit of $\log_2 c(k) = H(\theta)k - c'_2 \log_2 k + c_0$ on $[200, 1200]$ returns $c'_2 = 1.43$; the deviation from $\frac{3}{2}$ is the $O(1)$ oscillation, not a different exponent.

Corollary 7.6 (density; classical in content)**.** *The natural density of the union of climbing classes at depth k is $c(k)/2^k = 2^{-(1-H(\theta))k+o(k)}$, with $1 - H(\theta) = 0.0500444728\ldots$; equivalently, the greedy certificate tree of depth k certifies descent on a set of density $1 - c(k)/2^k \to 1$. This recovers, in certificate form, the density theorems of Terras and Everett [1, 2, 16].*

8 Sturmian structure of the frontier

For $j \geq 1$ let

$$\sigma_j = \lfloor (j+1)\theta \rfloor - \lfloor j\theta \rfloor \in \{0, 1\}$$

be the j-th letter of the *characteristic Sturmian word* of slope θ [26, Ch. 2].

Theorem C (Sturmian letter factorization). For a word $(b_1, \ldots, b_k)$ define the *excess walk* $d_j = e_j - \lfloor \theta j \rfloor - 1$. Then the word is climbing if and only if $d_j \geq 0$ for all $1 \leq j \leq k$, and the walk updates by

$$d_{j+1} - d_j = b_{j+1} - \sigma_j \in \begin{cases} \{0, +1\}, & \sigma_j = 0, \\ \{-1, 0\}, & \sigma_j = 1. \end{cases}$$

Hence $c(k)$ equals the number of non-negative paths of the σ-driven walk started from the forced first step $b_1 = 1$ (i.e. $d_1 = 0$).

Proof. Since θj is irrational for $j \geq 1$, $e_j > \theta j$ is equivalent to $e_j \geq \lfloor \theta j \rfloor + 1$, i.e. $d_j \geq 0$. The update is immediate from $e_{j+1} = e_j + b_{j+1}$ and the definition of σ_j. At $j = 1$: $\lfloor \theta \rfloor = 0$, so $d_1 = b_1 - 1 \geq 0$ forces $b_1 = 1$. □

For the block structure, write the continued fraction $\theta = [0; a_1, a_2, a_3, \ldots] = [0; 1, 1, 1, 2, 2, 3, 1, 5, 2, \ldots]$ and let $s_0 = 0$, $s_1 = 1$, $s_{n+1} = s_n^{a_{n+1}} s_{n-1}$ be the associated standard words; their lengths are the continued-fraction denominators $q_n = 1, 1, 2, 3, 8, 19, 65, 84, 485, 1054, \ldots$ of θ, and the characteristic word σ is their limit [26, Ch. 2]. Encode the two letter steps of Theorem C as infinite 0–1 transfer matrices acting on the state $d \in \{0, 1, 2, \ldots\}$, in the row-vector convention:

$$M(0)_{d,d'} = [d' = d] + [d' = d + 1], \qquad M(1)_{d,d'} = [d' = d - 1]_{d \geq 1} + [d' = d].$$

Products of k such matrices are banded (bandwidth $\leq k$), so all entries below are finite sums.

Theorem D (standard-word block recursion). Let $B_n = M(\sigma\text{-letters of } s_n)$ be the ordered product of letter matrices along the standard word s_n (row-vector convention: products compose in word order). Then

$$B_{n+1} = B_n^{a_{n+1}} B_{n-1} \qquad (n \geq 1),$$

and for every k,

$$c(k) = \left(e_0^\top \, M(\sigma_1) M(\sigma_2) \cdots M(\sigma_{k-1}) \right) \mathbf{1},$$

where e_0 is the coordinate vector of the state $d = 0$.

Proof. The map assigning to a finite word the ordered product of its letter matrices is a monoid morphism (in the row-vector convention, concatenation maps to product in the same order), so the matrix recursion is the image of the standard-word recursion $s_{n+1} = s_n^{a_{n+1}} s_{n-1}$. The formula for $c(k)$ is Theorem C: the matrix entry $(M(\sigma_1) \cdots M(\sigma_{k-1}))_{0,d}$ counts the walks from $d_1 = 0$ to $d_k = d$ that stay non-negative, and summing over d counts all climbing words. □

Remark 8.1 (numerics)**.** The identification of σ with the limit of the standard words was checked letter-by-letter to length 1054 ($= q_9$); the matrix recursion was verified exactly on truncation-exact corners and modulo the prime 3000017 for $n \leq 6$; and the matrix-product formula for $c(k)$ was checked against the direct dynamic program for $k \leq 200$. A two-letter statistic captures most of the fluctuation of $\log_2 c(k)$ around $H(\theta)k$: empirically $\mathbb{E}[\Delta\varphi \mid \sigma =$

$0] = +0.05004 \approx 1 - H(\theta)$ and $\mathbb{E}[\Delta\varphi \mid \sigma = 1] = -0.03022$ explain 82% of the variance, rising to 97.6% at the level of derived-word blocks — the fluctuation of the certificate count is governed by the first two levels of the continued-fraction hierarchy of $\log_2 3$.

Theorem E (non-regularity of the frontier). The language L of climbing words is not regular. More precisely, for $a, b \geq 1$,

$$1^a 0^b \in L \iff b \leq \left\lceil a \tfrac{1-\theta}{\theta} \right\rceil - 1,$$

and since the right-hand side is unbounded in a, the Myhill–Nerode equivalence has infinitely many classes.

Proof. For the word $1^a 0^b$ the prefix sums are $e_j - \theta j = j(1-\theta) > 0$ for $j \leq a$, and $e_{a+i} - \theta(a+i) = a - \theta(a+i)$ for $1 \leq i \leq b$, which is positive iff $i < a(1-\theta)/\theta$; as θ is irrational, equality is impossible, so the condition on b is $b \leq \lceil a(1-\theta)/\theta \rceil - 1$. Write $\phi(a)$ for this bound; ϕ is unbounded. If $a \neq a'$ with $\phi(a) < \phi(a')$, the word $0^{\phi(a')}$ separates 1ª from $1^{a'}$; hence infinitely many Myhill–Nerode classes, and L is not regular. □

Remark 8.2 (related work)**.** Balanced-word structures have independently surfaced on the cycle side of the problem: Fernández–Ibáñez [19] prove that Christoffel words — the finite relatives of standard Sturmian words — are, up to rotation, the unique maximizers of a cycle functional on binary words of fixed length and weight, yielding bounds on the minimum element of a hypothetical cycle. Their extremal role there and the driving role of the characteristic Sturmian word here (Theorems C–D) are distinct phenomena with a common source: the affine functional $3^e/2^j$ is extremized under balance constraints governed by the continued fraction of $\log_2 3$. In the machine-verified setting, Sharpe [21] excludes specific mechanical (Sturmian) parity itineraries of slopes above the critical line — slope $1/\sqrt{2}$ and an explicit metallic family — as itineraries of actual orbits; that is a statement about orbits, whereas Theorems C–E concern the frontier of certificate systems, whose slope is the critical θ itself.

Conjecture 8.3. *The sequence $c(k)$ is not P-recursive (holonomic). (Supporting evidence: an exhaustive search found no polynomial-coefficient recurrence of order ≤ 6 and degree ≤ 6, with the kernel computations confirmed modulo three independent primes; and the Sturmian driving of Theorem C ties the fine structure of $c(k)$ to the continued fraction of $\log_2 3$, which is not eventually periodic as far as it has been computed.)*

9 Local obstructions

The counting theorem bounds one specific proof system. It is natural to ask whether small modifications escape it: a bounded "syndrome correction" to the potential $\log_2 n$, or a hierarchy of finite-window constraints meant to exclude the climbing statistics. The following two results close the bounded-memory variants of both ideas; their scope limitations are stated explicitly and discussed in Section 11.

Theorem F (no bounded dyadic one-step Lyapunov function). Let $d \geq 1$, $\psi : \mathbb{Z}/2^d\mathbb{Z} \to \mathbb{R}$, $\varepsilon > 0$, and set $\varphi(n) = \log_2 n + \psi(n \bmod 2^d)$. Then the one-step descent property

$$\varphi(T(n)) \leq \varphi(n) - \varepsilon \qquad \text{for all } n > 1$$

fails. In fact it fails on the single class $n \equiv -1 \pmod{2^{d+1}}$: there the syndrome is a fixed point and the step is expanding.

Proof. Let $n = 2^{d+1}t - 1$, $t \geq 1$. Then n is odd, $n \equiv -1 \pmod{2^d}$, and by Lemma 4.2 (with $m = d + 1$) $T(n) = 3 \cdot 2^d t - 1 \equiv -1 \pmod{2^d}$. Hence $\psi(T(n) \bmod 2^d) = \psi(n \bmod 2^d)$ and

$$\varphi(T(n)) - \varphi(n) = \log_2 \frac{3n+1}{2n} > \log_2 \frac{3}{2} > 0,$$

contradicting descent by any $\varepsilon > 0$. □

Remark 9.1 (robustness of the obstruction)**.** Theorem F is stated for the all-ones syndrome because one class suffices, but the obstruction is much fatter: in the natural linear-programming relaxation over the syndrome transition graph, every shift-invariant measure on parity sequences with odd-step frequency $\rho > \theta$ has positive drift $\rho \log_2 3 - 1 > 0$, and these measures form a non-empty compact convex set. Any per-step, bounded-memory dyadic correction thus fails robustly, not by a degenerate coincidence. By contrast, deleting the states with two trailing ones makes the maximum mean cycle negative: an explicit finite Lyapunov function exists for orbits that never visit $n \equiv 3 \pmod 4$; compare Corollary 9.3.

Theorem G (cylinder-local realization). Fix a window length $w \geq 1$ and a syndrome depth $d \geq 1$. There exist infinitely many n (namely all $n = 2^{w+d+1}t - 1$, $t \geq 1$) whose first w compressed steps form a strictly increasing orbit segment with all steps odd, unit 2-adic valuations ($v_2(3n_j + 1) = 1$), and constant all-ones syndrome $n_j \equiv -1 \pmod{2^d}$ for $0 \leq j \leq w$. Consequently, any family of constraints on length-w windows of (parities, valuations, depth-d syndromes) that is valid for all non-descending integer orbit segments is satisfied by the all-ones statistics: finite-window constraint hierarchies cannot exclude climbing statistics.

Proof. Apply Lemma 4.2 with $m = w + d + 1$: for $j \leq w$ the values $n_j = 3^j 2^{m-j} t - 1$ satisfy $n_j \equiv -1 \pmod{2^{m-j}}$ and $m - j \geq d + 1$, so all listed statistics of the window coincide with those of the all-ones cylinder, while the segment is a genuine orbit of a positive integer and is strictly increasing (hence non-descending). □

Remark 9.2 (scope)**.** Theorem G is a $\forall w \, \exists n$ statement: for each window length some integer realizes the climbing statistics. It does not produce a single n realizing them for all w (that would be a divergent orbit). Its force is contrapositive: any proof method whose reach is exhausted by constraints on bounded windows of bounded-depth dyadic data cannot prove the conjecture, since such constraints hold equally in the 2-adic completion, where the all-ones orbit ($-1 \mapsto -1$) exists. Local statistics see only $\mathbb{Z}_2$, not $\mathbb{N}$.

Corollary 9.3 (forced visits to 3 mod 4)**.** *If $n > 3^k$ and $T^j(n) \geq n$ for all $j \leq k$, then the orbit segment visits $\{m \equiv 3 \pmod 4\}$ at least $(2\theta - 1)k - 1$ times among $j = 0, \ldots, k-1$; the frequency is at least $2\theta - 1 - O(1/k)$, with $2\theta - 1 = 0.2618595\ldots$*

Proof. Non-descent through depth k forces the climbing condition $e_j > \theta j$ for all $j \leq k$: if some prefix had $3^{e_j} < 2^j$, then by Lemma 2.1(b) descent $T^j(n) < n$ would hold whenever $n(2^j - 3^{e_j}) > \beta_j$, which is guaranteed by $n > 3^k \geq \beta_j$ since $2^j - 3^{e_j} \geq 1$. An odd m has $T(m)$ odd iff $3m + 1 \equiv 2 \pmod 4$ iff $m \equiv 3 \pmod 4$. Thus visits to 3 mod 4 within the window are exactly the adjacent pairs $b_j = b_{j+1} = 1$. A binary word of length k with e ones has at least $e - (k - e + 1) = 2e - k - 1$ such pairs (the ones form at most $k - e + 1$ blocks). With $e \geq \theta k$ this gives $(2\theta - 1)k - 1$. □

10 Verified computational artifacts

10.1 The depth-20 certificate system

A greedy certificate tree of dyadic depth 20 was generated and independently verified. It consists of 4404 certificates whose atoms, together with the $c(20) = 27328$ climbing classes mod 2^{20}, form an exact partition of $\mathbb{Z}/2^{20}\mathbb{Z}$; the certified classes have natural density $1 - 27328/2^{20} = 97.39\%$, and the single exceptional integer across the entire system is $n = 1$ (the endpoint of the trivial cycle, for which descent is not required). Each certificate is stored as the full tuple $(k,\ \mathrm{mod},\ r,\ e,\ \beta,\ \mathrm{exceptions})$ in a JSON artifact.

The verifier is deliberately independent of the generator: it re-derives the parity vectors and affine data symbolically from scratch (Lemma 2.1), checks exact cover and disjointness of the atoms, re-checks the descent inequality $n(2^k - 3^e) > \beta$ at the minimal element of each class (which suffices for the whole class by monotonicity of the affine form), and confirms the climbing condition for each unresolved class. All arithmetic is exact (arbitrary-precision integers).

10.2 Machine verification of the core lemmas

The arithmetic cores of Theorems A and A′ are verified in Lean 4, compiled with core Lean only (no external libraries), toolchain pinned to `leanprover/lean4:v4.32.2`, with no `sorry`:

- `Barrier.lean`: Lemma 4.2 as four theorems (`climb`, `climb_mod`, `climb_ge`, `barrier`); axioms exactly `[propext, Quot.sound]`.
- `Collapse.lean`: the affine identity of Lemma 2.1(b) for the full compressed map (`affine`), the remainder bound $\beta_j + 2^j \le 3^j$ (`beta_bound`), both halves of the collapse lemma (Lemma 6.2) as `climbing_no_descent` ($2^j < 3^{e_j} \Rightarrow T^j(n) > n$ for every $n \ge 1$) and `descent_forced` ($3^{e_j} < 2^j$ and $n > 3^j \Rightarrow T^j(n) < n$), and the arithmetic core of the persistence lemma (Lemma 6.3) as `climb_extend`; axioms within `[propext, Classical.choice, Quot.sound]`, the standard Lean axioms.
- `Rotation.lean`: the rotation lemma (Lemma 7.1) in the integer cross-multiplied form of Remark 7.2 (`rotation`, built on a verified last-minimum scan `am` with its minimality and strictness lemmas); axioms within `[propext, Classical.choice, Quot.sound]`. The correspondence between the cross-multiplied conclusion and the literal climbing of the rotated word was additionally cross-checked exhaustively for all words of length ≤ 14.
- `Counting.lean`: *Theorem A itself*, in an abstract certificate-system formulation. A system is abstracted as an index type I with depth $\mathrm{dep}\, i$, atom $\mathrm{atom}\, i$, and a coverage predicate, subject to exactly the three hypotheses used in the proof: containment in the dyadic atom, descent at the certificate's own depth on every covered integer, and coverage of all $n \ge N_0$ (which absorbs exceptional sets, odd-modulus refinements, and the finite directly verified set). The conclusion (`counting_lower_bound`) is constructive: an *injection* from the climbing residues mod 2^K into $\{i : \mathrm{dep}\, i > K\}$ — the statement "at least $c(K)$ certificates of depth $> K$" rendered without cardinality machinery — together with the corollary `no_finite_system` ($\forall K\, \exists i,\ \mathrm{dep}\, i > K$, witnessed by the all-ones class). The key lemma is `odds_mod`: the first j parities depend only on n mod 2^j (the part of Lemma 2.1 the proof uses). Axioms within `[propext, Classical.choice, Quot.sound]`.

- `TheoremB.lean` (separate Lake project, Lean `v4.33.1` with Mathlib pinned at the matching release): *the two-sided sandwich of Theorem B.* Words are `List.Vector Bool k`, the climbing count $c(k)$ is the cardinality of the climbing words, and $e_0 = \min\{e : 2^k < 3^e\}$; the theorem `theoremB` states $\binom{k}{e_0} \le k \cdot c(k)$ and $c(k) \le (k+1)\binom{k}{e_0}$ for $k \ge 1$. The lower bound composes the integer rotation theorem with the list identities for prefix counts of `List.rotate` and the fiber bound of the map "word $\mapsto$ its climbing rotation" (`Finset.card_le_mul_card_image`); the upper bound uses $2e_0 > k$ and the monotone decrease of binomial coefficients above the middle; the identification of $\#\{\text{words with } e \text{ ones}\}$ with $\binom{k}{e}$ goes through indicator vectors of `Finset.powersetCard`. Axioms within `[propext, Classical.choice, Quot.sound]`; no `sorry`.

- `CountingCorollary.lean` (same Lake project): *the combinatorial core of Corollary 6.5.* With T the compressed map and `stays m n` the predicate "the first m iterates stay $\ge n$" (which contains $\mathcal{E} \cap (2^m, 2^{m+1}]$), `growth_ineq` proves the integer form $2^{m(e+1)} \le (3 \cdot 2^m + 1)^e$ of $e \ge m\theta_M$ for $n > 2^m$, `parityset_injective` proves the Terras injectivity of $n \mapsto \{i < m : T^i n \text{ odd}\}$ on $(2^m, 2^{m+1}]$ via the affine difference identity $2^j(T^j n - T^j n') = 3^{e_j}(n - n')$, and `count_le` concludes $\#\{n \in (2^m, 2^{m+1}] : \texttt{stays}\} \le \sum_e \binom{m}{e}$ over the admissible e. Only the entropy estimate $\sum_{e \ge pm} \binom{m}{e} \le 2^{mH(p)}$ and the passage from the integer inequality to θ_M are left to the text. Axioms within `[propext, Classical.choice, Quot.sound]`; no `sorry`.

Of the main results, Theorem A (modulo the trivial reading of an injection as a statement about distinct classes), the binomial sandwich of Theorem B, and the combinatorial core of Corollary 6.5 are thus machine-verified, while the entropy limit in Theorem B (Fekete and Stirling), the sharp growth of Theorem 7.4, and the remaining steps of A′, as well as Theorems C–G, are verified by conventional proof plus the independent computations described above. The three core files use no library at all and can be checked in seconds; the Mathlib project requires a Mathlib cache. Both toolchains used (`v4.32.2`, released 2026-07-28, and `v4.33.1`) postdate the fix for the Lean kernel soundness issue #14576 of July 2026, which in any case was reachable only through metaprogrammed kernel submissions; the artifacts use no metaprogramming.

10.3 Reproducibility and data availability

All computations use exact integer (or exact rational) arithmetic. The supplementary bundle contains the certificate artifact, the Lean file with its pinned toolchain, the independent verifier, and one self-contained script per computational claim of this paper, each printing the quantity it certifies; a README lists the exact commands and expected outputs. SHA-256 hashes of the frozen artifacts:

artifact	SHA-256
`certificates.json`	`d32a24e4b34eb33a0aa8f01406354ae2`
	`2ddd65a7329b3d3309197150ad3d78fc`
`Barrier.lean`	`c416508cda41b0df105bae91801d70c3`
	`7a3e2f65c0251bae1c90683492f4ec7d`
`Collapse.lean`	`08191fc7b0114d8504a9c930ae911ad7`
	`075f33d05a38b6e26612bad2cf50b2f3`
`Rotation.lean`	`20f91338f3d623a496ce1d0a6b1ff775`
	`75f14921756e17dc13ab48bd5863b97a`
`Counting.lean`	`09619f3b7862934a14ef7fb9ed9d4646`
	`a50ea425a4622f819acf27bded5a4396`
`TheoremB.lean`	`0b2f7f732e6e42dc45624531a3e5c05a`
`CountingCorollary.lean`	`eb05554941149f2930d3579a984106b3`
	`8847c88912e3ed3321a716dd8fd3879a`

11 Discussion: what is and is not blocked

Theorems A and A′ block the finite-horizon descent-certificate strategy in full generality — any mixture of depths, congruence scopes of arbitrary moduli, disjunctions, adaptive stopping, and finite exceptions — and Theorems F–G block its bounded-memory Lyapunov and finite-window relaxations. It is equally important to state what is *not* blocked.

Congruence information is exhausted. By Theorem A′, no repackaging of congruence conditions on the starting point — dyadic, 3-adic, or mixed — escapes the count: the certificate frontier is canonical, and its cost $2^{H(\theta)k}$ is invariant. A proof of the conjecture by descent certificates would need scopes that are not eventually periodic sets: sets defined by inequalities on orbit values, or by information of unbounded window length. The same collapse yields Corollary 6.5 — unconditionally at most $N^{0.950}$ potential counterexamples up to N — and, since the congruence necessary conditions are attained exactly, improving that exponent requires non-shadow information.

Unbounded memory and Lyapunov potentials. Theorem F excludes only per-step, bounded-memory dyadic corrections, and Theorem A′ closes variable stopping for the potential $\log_2 n$ itself. Lyapunov schemes with unbounded memory, or with potentials other than $\log_2 n$ certified at variable times, are untouched — though we note that the certificate tree itself is such an unbounded-memory object, and Theorem A prices its frontier at $2^{H(\theta)k}$ classes per depth k.

Non-dyadic information along the orbit. Theorems F–G are 2-adic, and Theorem A′ blocks 3-adic conditions only as congruence scopes on the initial point. Methods that consume 3-adic equidistribution *along the orbit of an ensemble* — in particular the Fourier-analytic route of Tao [5], which transports the initial distribution of an ensemble rather than finite windows of a single orbit — remain outside the scope of all our obstructions. We analyze the quantitative 3-adic mixing problem in a companion paper.

The cycle route. Diophantine bounds on nontrivial cycles (linear forms in logarithms combined with verified lower bounds on cycle elements [7, 8, 9, 6]) address the periodicity half of the conjecture and are independent of everything here; the appendix records the standard bound in the form we use.

Finally, the barrier perspective suggests reading the classical density theorems not as progress toward the conjecture but as exact complexity measurements of a proof system: the certified density $1 - 2^{-(1-H(\theta))k}$ and the certificate count $2^{H(\theta)k}$ are two faces of the same entropy, and

the fine structure of the frontier is governed by the continued fraction of $\log_2 3$ (Theorems C–E) — the same Diophantine data that governs the cycle route. We take this coincidence as evidence that the difficulty quantified here is intrinsic to the problem and not to one formalization.

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The author reports there are no competing interests to declare.

Declaration of generative AI use

Generative AI (Anthropic's Claude) was used, under the author's direction, to assist in drafting and revising the manuscript, in developing and checking the computational scripts and the Lean 4 proof files, and in searching the literature. All definitions, theorems, and proofs were reviewed by the author, the computational claims are backed by the verifiable artifacts described in this article, and the author takes full responsibility for its content.

Data availability statement

All code and data supporting this study — the certificate artifact, the Lean 4 proof with pinned toolchain, the independent verifier, and one self-contained exact-arithmetic script per computational claim, with a claim-to-script map and SHA-256 manifest — are openly available in the reproduction repository `collatz-barrier-repro` (https://github.com/WakauraH/collatz-barrier-repro); a DOI-stamped archival snapshot of the exact version used here will be deposited upon acceptance. The same material is included as the supplementary bundle of this submission.

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A The standard cycle bound (reproduction)

For completeness we record the cycle bound used in Section 11; the method is that of Eliahou [7] with the current verification bound, and no novelty is claimed. Let a nontrivial cycle of the odd-to-odd accelerated map have odd elements $m_1, \dots, m_K$ with 2-adic valuations summing to A. Multiplying the relations $m_{j+1}2^{a_j} = 3m_j + 1$ around the cycle gives $2^A = \prod_j (3 + 1/m_j)$, hence with $M = \min_j m_j$,

$$0 < A - K\log_2 3 \le \frac{K}{3M \ln 2}.$$

With $M > 2^{68}$ [6] the right side is below $K \cdot 2^{-69}$, forcing $A = \lceil K \log_2 3 \rceil$ and making $\|K \log_2 3\|^+$ (the distance to the integer *above*) at most $K \cdot 2^{-69}$. Running the one-sided ladder of best approximations of $\log_2 3$ (mediant steps between convergents, compared by exact integer powers 2^p vs 3^q) shows the smallest such K satisfies

$$K \;\ge\; 8{,}963{,}457{,}696,$$

so any nontrivial cycle has more than 2.3×10^{10} steps of the map C. Stronger bounds via m-cycle theory are known [8], and high cycles have recently been eliminated altogether [20].